SIMATS ENGINEERING

CSA43 - INTERNET PROGRAMMING

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Course Code & Title	CSA43 - Internet Programming

EXPERIMENT 1: LIBRARY MANAGEMENT SYSTEM

Aim:

To design and develop a Library Management web application using HTML, CSS and JavaScript that allows the librarian/admin to add, update, issue and return books and calculate fines, and allows users to register, log in, search books, view borrow history, check fines and download study material.

Code (HTML, CSS and JavaScript):

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>City Library Management System</title>
<style>
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: 'Segoe UI', Arial, sans-serif;
}
body {
  background: #eef4ff;
  color: #1e293b;
}
header {
  background: linear-gradient(120deg, #1e3a8a, #3b82f6);
  color: white;
  padding: 30px 20px;
```

```
    text-align: center;
}
header h1 {
    font-size: 30px;
}
header p {
    margin-top: 8px;
    color: #dbeafe;
}
.container {
    max-width: 900px;
    margin: 30px auto;
    padding: 0 20px;
}

.card {
    background: white;
    padding: 25px;
    margin-bottom: 25px;
    border-radius: 15px;
    box-shadow: 0 5px 20px rgba(30, 58, 138, 0.1);
    transition: 0.3s;
}
.card:hover {
    transform: translateY(-3px);
    box-shadow: 0 8px 25px rgba(30, 58, 138, 0.15);
}
h2 {
    color: #1e3a8a;
    margin-bottom: 18px;
    border-left: 5px solid #3b82f6;
    padding-left: 12px;
}
input {
    width: 100%;
    padding: 12px;
    margin: 8px 0 15px;
    border: 1px solid #cbd5e1;
    border-radius: 8px;
    outline: none;
}
input:focus {
    border-color: #3b82f6;
    box-shadow: 0 0 5px #bfdbfe;
}
button {
    background: #2563eb;
    color: white;
    border: none;
    padding: 11px 20px;
    border-radius: 8px;
    cursor: pointer;
    font-weight: bold;
    transition: 0.3s;
```

```
}
button:hover {
  background: #1e40af;
  transform: scale(1.03);
}
.return {
  background: #16a34a;
}
.return:hover {
  background: #15803d;
}

table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 15px;
  overflow: hidden;
}
th {
  background: #1e3a8a;
  color: white;
  padding: 13px;
  text-align: left;
}
td {
  padding: 12px;
  border-bottom: 1px solid #e2e8f0;
}
tr:hover td {
  background: #eff6ff;
}
.status {
  color: #15803d;
  font-weight: bold;
}
footer {
  background: #1e3a8a;
  color: white;
  text-align: center;
  padding: 18px;
  margin-top: 30px;
}
#result {
  margin-top: 15px;
  font-size: 18px;
  font-weight: bold;
  color: #dc2626;
}
@media (max-width: 600px) {
  header h1 {
    font-size: 23px;
  }
  .card {
    padding: 18px;
```

```
  }
  table {
    font-size: 12px;
  }
  th, td {
    padding: 8px;
  }
}
</style>
</head>
<body>
<header>
  <h1>📚 City Library Management System</h1>
  <p>Your Smart and Simple Digital Library</p>
</header>

<div class="container">
  <!-- Add Book -->
  <div class="card">
    <h2>📖 Add New Book</h2>
    <input id="title" placeholder="Enter Book Title">
    <input id="author" placeholder="Enter Author Name">
    <button onclick="addBook()">+ Add Book</button>
  </div>
  <!-- Search and Issue / Return -->
  <div class="card">
    <h2>🔍 Search & Issue / Return</h2>
    <input id="search" placeholder="Search by title or author..." onkeyup="filterBooks()">
    <table id="books">
      <tr>
        <th>Book Title</th>
        <th>Author</th>
        <th>Status</th>
        <th>Action</th>
      </tr>
      <tr>
        <td>Internet Programming</td>
        <td>Deitel</td>
        <td class="status">Available</td>
        <td><button onclick="toggleBook(this)">Issue</button></td>
      </tr>
      <tr>
        <td>Machine Learning</td>
        <td>Tom Mitchell</td>
        <td class="status">Available</td>
        <td><button onclick="toggleBook(this)">Issue</button></td>
      </tr>
      <tr>
        <td>Database Systems</td>
        <td>Navathe</td>
        <td class="status">Available</td>
        <td><button onclick="toggleBook(this)">Issue</button></td>
      </tr>
    </table>
```

```
    </div>

    <!-- Fine Calculator -->
    <div class="card">
      <h2>💰 Fine Calculator</h2>

      <input type="number" id="days" min="0" placeholder="Enter number of late days">
      <button onclick="calculateFine()">Calculate Fine</button>
      <p id="result"></p>
    </div>
  </div>
  <footer>
    © 2026 City Library | Developed by Kirthanaa
  </footer>
  <script>
  function addBook() {
    let title = document.getElementById("title").value.trim();
    let author = document.getElementById("author").value.trim();
    if (title === "" || author === "") {
      alert("Please enter book title and author");
      return;
    }
    let table = document.getElementById("books");
    let row = table.insertRow();

    row.innerHTML = `
      <td>${title}</td>
      <td>${author}</td>
      <td class="status">Available</td>
      <td><button onclick="toggleBook(this)">Issue</button></td>
    `;
    document.getElementById("title").value = "";
    document.getElementById("author").value = "";
  }
  function toggleBook(button) {
    let row = button.parentElement.parentElement;
    let status = row.cells[2];

    if (status.innerText === "Available") {
      status.innerText = "Issued";
      status.style.color = "#dc2626";
      button.innerText = "Return";
      button.classList.add("return");
    } else {
      status.innerText = "Available";
      status.style.color = "#15803d";
      button.innerText = "Issue";
      button.classList.remove("return");
    }
  }
  function filterBooks() {
    let search = document.getElementById("search").value.toLowerCase();
    let rows = document.querySelectorAll("#books tr");
```

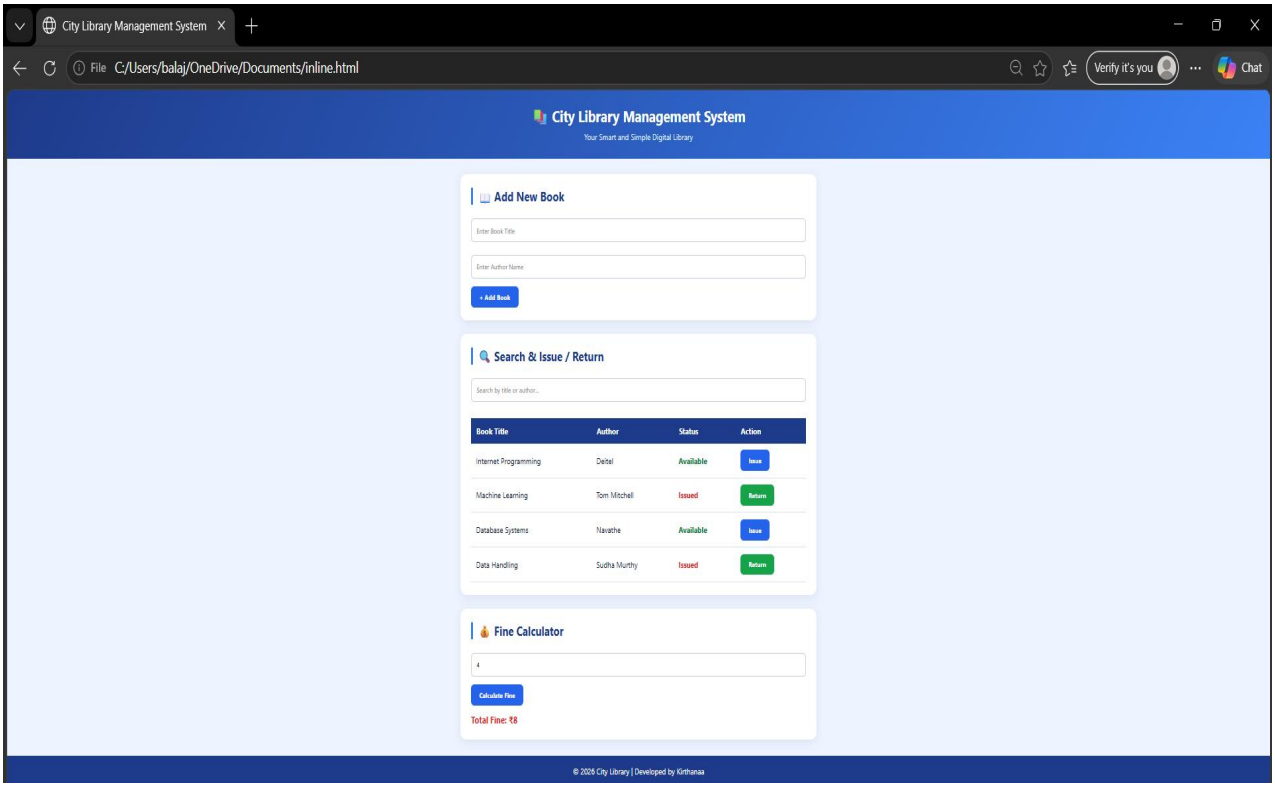
```
  rows.forEach((row, index) => {
    if (index === 0) return;

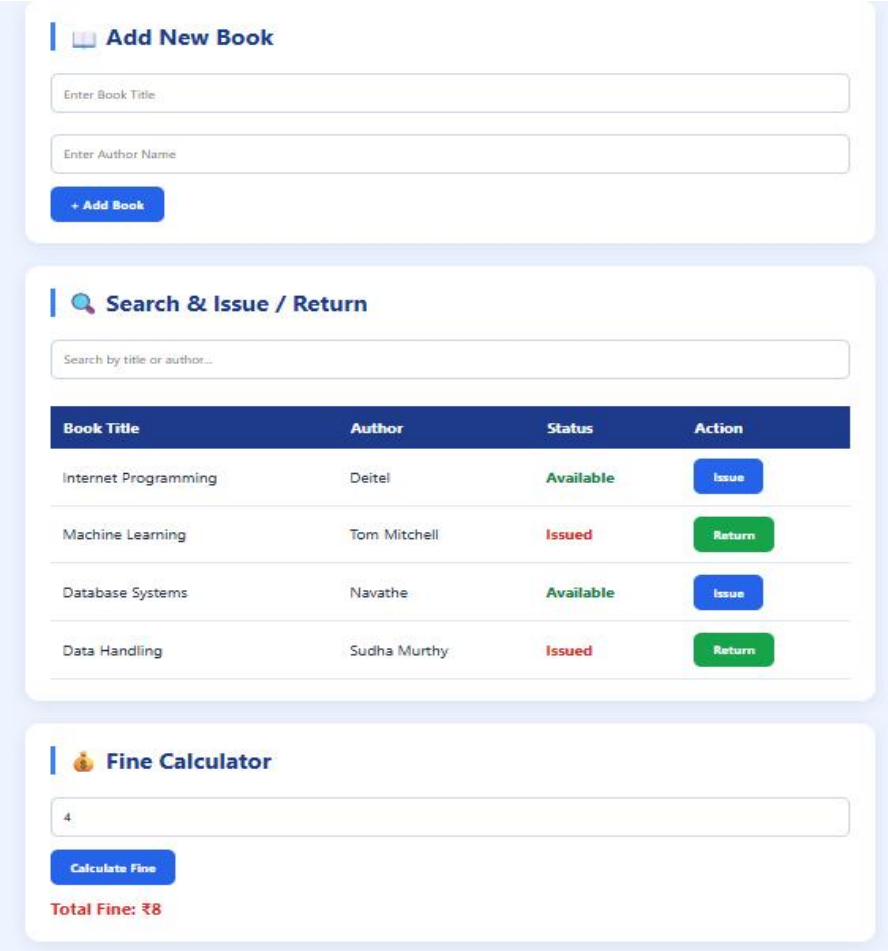
    row.style.display =
      row.innerText.toLowerCase().includes(search) ? "" : "none";
  });
}

function calculateFine() {
  let days = document.getElementById("days").value;
  if (days === "" || Number(days) < 0) {
    alert("Please enter valid late days");
    return;
  }

  let fine = Number(days) * 2;
  document.getElementById("result").innerText =
    "Total Fine: ₹" + fine;
}
</script>
</body>
</html>
```

OUTPUT:





Result:

Thus the Library Management System web application was designed, developed and executed successfully.

EXPERIMENT 2: ONLINE HOTEL ROOM BOOKING SYSTEM

Aim:

To develop a fully responsive online hotel room booking website using HTML, CSS and JavaScript that lets users search hotels by destination, dates and guests, check room availability, book/modify/cancel bookings, and lets the administrator manage room types, prices, bookings and static information pages.

Code (HTML, CSS and JavaScript):

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>StayEasy Hotel Booking</title>
<style>
*{box-sizing:border-box;margin:0;padding:0;font-family:'Segoe UI',Arial,sans-serif}
body{background:#fff7ed;color:#292524}
header{background:linear-gradient(120deg,#7c2d12,#ea580c);color:white;padding:22px;text-align:center}
header h1{font-size:28px}
.hero{background:linear-gradient(135deg,#ea580c,#9a3412);color:white;text-align:center;padding:40px 20px}
.hero h2{font-size:30px;margin-bottom:15px}
.search{background:white;max-width:750px;margin:20px auto;padding:20px;border-radius:12px;display:flex;gap:10px;flex-wrap:wrap}
input{flex:1;min-width:150px;padding:12px;border:1px solid #ddd;border-radius:8px}
button{background:#ea580c;color:white;border:none;padding:12px 18px;border-radius:8px;cursor:pointer;font-weight:bold}
button:hover{background:#9a3412}
.container{max-width:1000px;margin:30px auto;padding:0 20px}
h2{color:#9a3412;margin-bottom:20px}
.rooms{display:grid;grid-template-columns:repeat(auto-fit,minmax(210px,1fr));gap:20px}
.room{background:white;border-radius:12px;overflow:hidden;box-shadow:0 4px 15px #ddd;transition:0.3s}
.room:hover{transform:translateY(-5px)}
.img{background:linear-gradient(135deg,#fb923c,#9a3412);color:white;padding:30px;text-align:center;font-size:40px}
.details{padding:18px}
.details p{color:#78716c;margin:10px 0}
.price{color:#ea580c;font-size:22px;font-weight:bold}
.booking{background:white;padding:20px;margin-top:25px;border-radius:12px;box-shadow:0 4px 15px #ddd}
#result{margin-top:15px;color:#15803d;font-weight:bold}
footer{background:#7c2d12;color:white;text-align:center;padding:18px;margin-top:30px}
@media(max-width:600px){.search{flex-direction:column}}
</style>
</head>
<body>
<header>
```

```
<h1>🏨 StayEasy Hotels</h1>
<p>Your Comfortable Stay, Just a Click Away</p>
</header>
<div class="hero">
<h2>Find Your Perfect Stay</h2>
<p>Discover comfortable rooms at affordable prices</p>
<div class="search">
<input id="destination" placeholder="Enter Destination">
<input id="checkin" type="date">
<input id="checkout" type="date">
<button onclick="checkAvailability()">Search Rooms</button>
</div>
<p id="availability"></p>
</div>
<div class="container">
<h2>🛏 Available Room Types</h2>
<div class="rooms">
<div class="room">
<div class="img">⭐ </div>
<div class="details">
<h3>Standard Room</h3>
<p>WiFi, TV, AC</p>
<div class="price">₹1,800/night</div>
<br>
<button onclick="book('Standard Room',1800)">Book Now</button>
</div>
</div>
<div class="room">
<div class="img">⭐ ⭐ </div>
<div class="details">
<h3>Deluxe Room</h3>
<p>Sea View, Breakfast</p>
<div class="price">₹3,200/night</div>
<br>
<button onclick="book('Deluxe Room',3200)">Book Now</button>
</div>
</div>
```

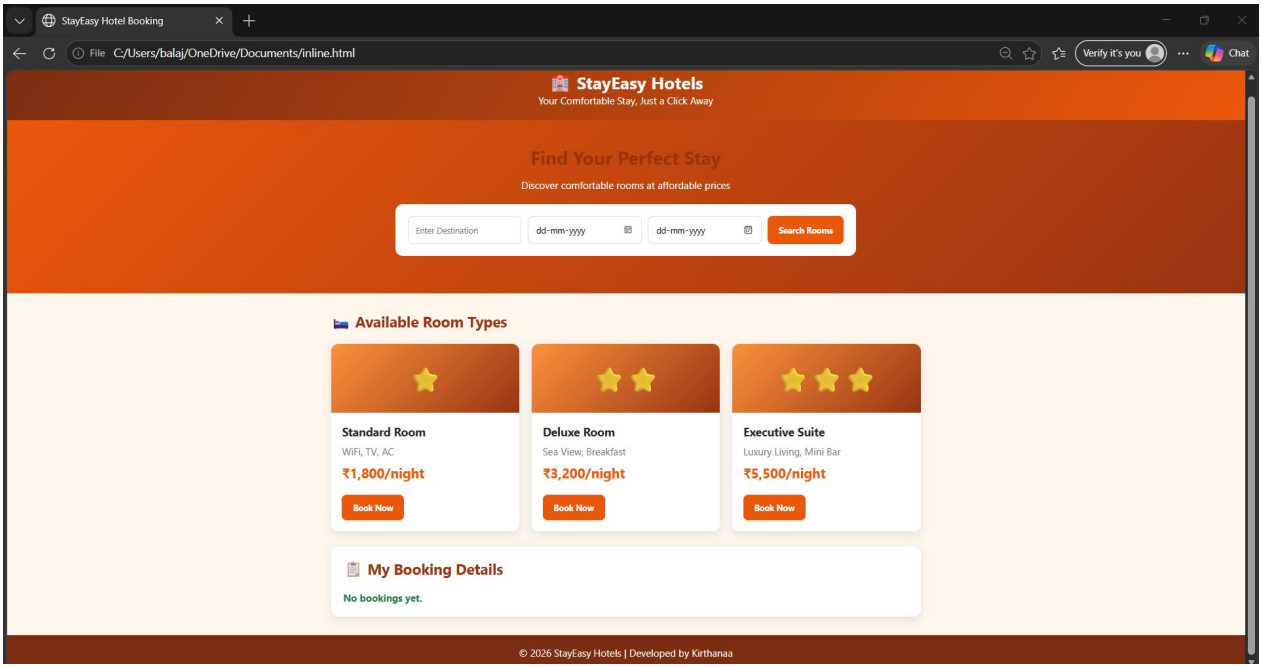
```
<div class="room">
<div class="img">⭐ ⭐ ⭐ </div>
<div class="details">
<h3>Executive Suite</h3>
<p>Luxury Living, Mini Bar</p>
<div class="price">₹5,500/night</div>
<br>
<button onclick="book('Executive Suite',5500)">Book Now</button>
</div>
</div>
</div>
<div class="booking">
<h2> 📋 My Booking Details</h2>
<p id="result">No bookings yet.</p>
</div>
</div>
<footer>
© 2026 StayEasy Hotels | Developed by Kirthanaa
</footer>
<script>
function checkAvailability(){
let destination=document.getElementById("destination").value;
let checkin=document.getElementById("checkin").value;
let checkout=document.getElementById("checkout").value;
if(destination===""||checkin===""||checkout===""){
alert("Please enter all search details");
return;
}
if(checkout<=checkin){
alert("Check-out date must be after check-in date");
return;
}
document.getElementById("availability").innerText="Rooms available in "+destination+"!";
}
function book(room,price){
document.getElementById("result").innerText=room+" booked successfully! Price: ₹"+price+" per night.";
}
```

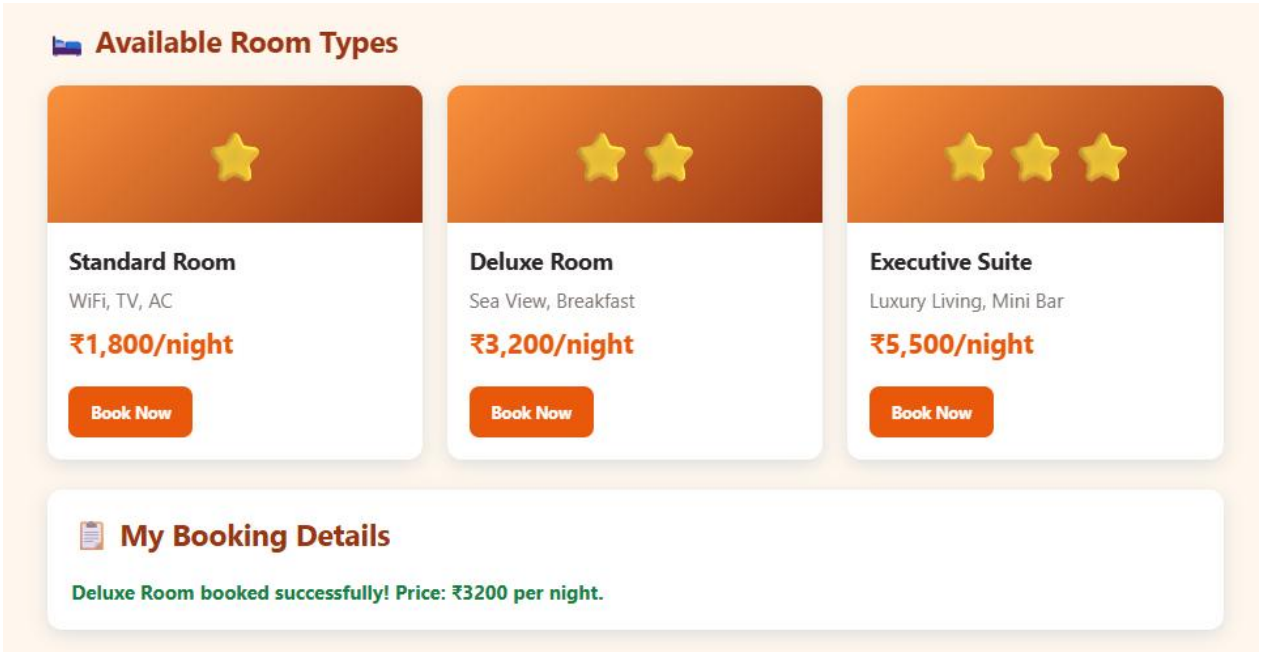
</script>

</body>

</html>

OUTPUT:





Result:

Thus the Online Hotel Room Booking System web application was designed, developed and executed successfully.

EXPERIMENT 3: ONLINE BUS TICKET BOOKING SYSTEM

Aim:

To design and implement a mobile-responsive online bus ticket booking website using HTML, CSS and JavaScript that allows users to search buses by route and date, select seats, enter passenger details, book/cancel tickets and post comments, and allows the admin to manage buses, users and reports.

Code (HTML, CSS and JavaScript):

```
<!DOCTYPE html>
<html lang="en">
<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">

<title>RedRoute Bus Booking</title>

<style>
*{box-sizing:border-box;margin:0;padding:0;font-family:Arial}
body{background:#f4f6fb;color:#333}
header{background:#be123c;color:white;padding:18px}
.hero{text-align:center;background:#fb7185;color:white;padding:30px}
.search,.bus,.box{background:white;padding:15px;margin:15px auto;border-radius:10px;max-width:850px}
.search input,.search select{padding:10px;margin:5px;border:1px solid #ddd}
button{background:#be123c;color:white;border:0;padding:10px;border-radius:6px;cursor:pointer}
.bus{display:flex;justify-content:space-between;align-items:center}
.bus h3{margin-bottom:8px}
.price{color:#be123c;font-size:20px;font-weight:bold}
.seats{display:grid;grid-template-columns:repeat(4,1fr);gap:8px;margin-top:15px}
.seat{background:#dcfce7;padding:10px;text-align:center;cursor:pointer;border-radius:5px}
.seat.booked{background:#ddd;cursor:not-allowed}
.seat.selected{background:#be123c;color:white}
footer{text-align:center;background:#881337;color:white;padding:15px}
</style>

</head>
<body>

<header><h2>🚌 RedRoute Bus Tickets</h2></header>
<div class="hero"><h2>Book Your Bus Ticket</h2>
<div class="search">
<select><option>Chennai</option><option>Bangalore</option></select>
<select><option>Coimbatore</option><option>Madurai</option></select>
<input type="date"><button onclick="alert('Searching buses...')">Search</button>
</div></div>
```

```
<div class="bus"><div><h3>Orange Travels</h3>Chennai → Coimbatore<br>10:30 PM - 05:30 AM</div><div><p class="price">₹950</p><button>Select</button></div></div>
<div class="bus"><div><h3>KPN Express</h3>Chennai → Coimbatore<br>09:45 PM - 05:00 AM</div><div><p class="price">₹1,150</p><button>Select</button></div></div>

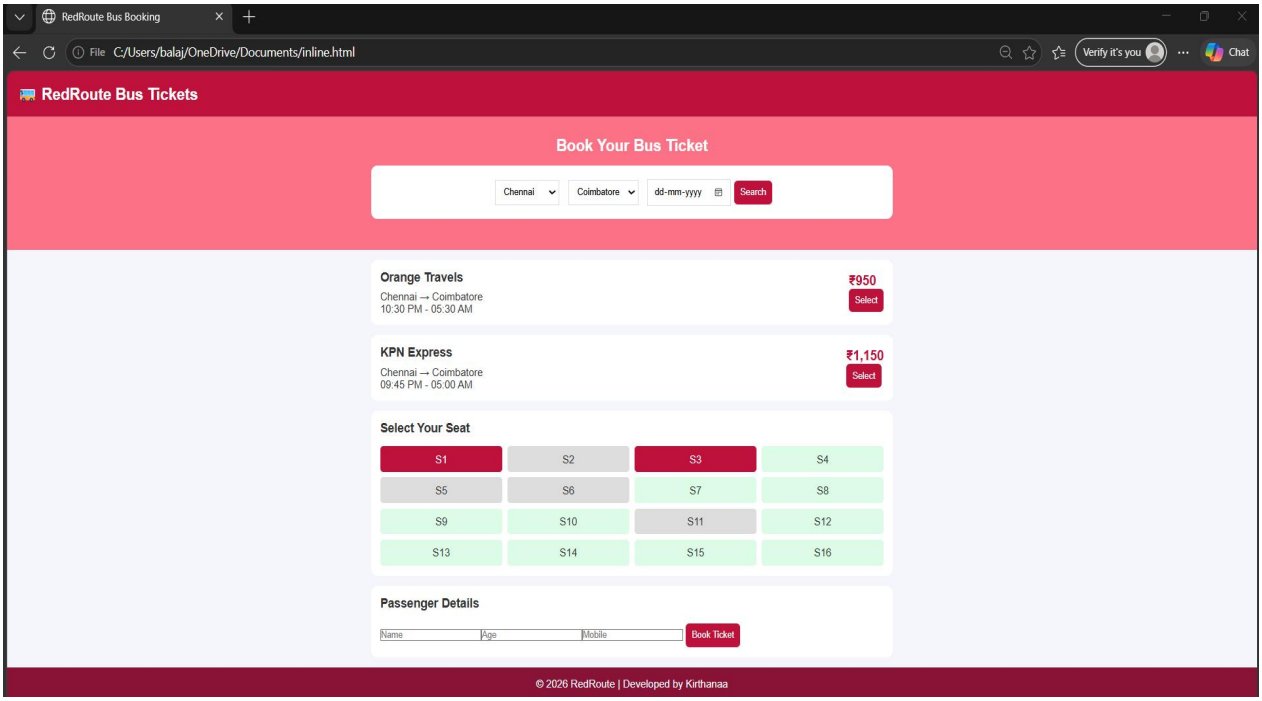
<div class="box"><h3>Select Your Seat</h3><div class="seats" id="seats"></div></div>
<div class="box"><h3>Passenger Details</h3><br>
<input placeholder="Name"><input placeholder="Age"><input placeholder="Mobile">
<button onclick="alert('Booking successful!')">Book Ticket</button>

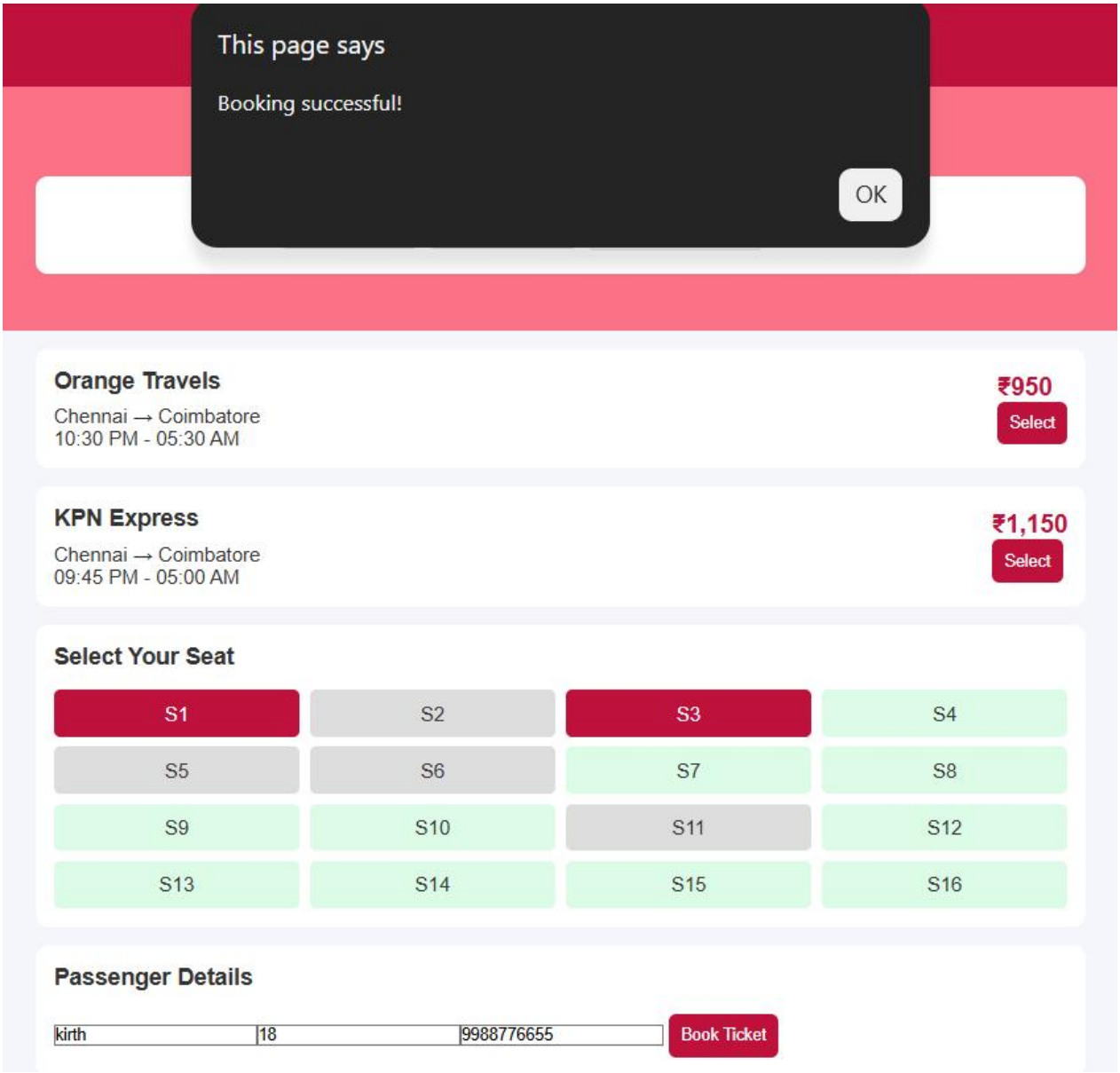
</div>
<footer>© 2026 RedRoute | Developed by Kirthanaa</footer>

<script>
var seats=document.getElementById("seats");
for(var i=1;i<=16;i++){
var s=document.createElement("div");
s.className="seat"+([2,5,6,11].includes(i)?" booked":"");
s.innerText="S"+i;
if(!s.classList.contains("booked"))
s.onclick=function(){this.classList.toggle("selected")};
seats.appendChild(s);
}

</script>
</body>
</html>
```

Output:





Result:

Thus the Online Bus Ticket Booking System web application was designed, developed and executed successfully.

EXPERIMENT 4: ONLINE TRAIN RESERVATION SYSTEM

Aim:

To develop a responsive online train reservation website using HTML, CSS and JavaScript that lets passengers search trains, check seat availability, book tickets with online/offline payment, check PNR status and cancel tickets with refund, while the admin maintains train, fare and payment details.

Code (HTML, CSS and JavaScript):

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">

<title>RailConnect Train Reservation</title>

<style>
*{box-sizing:border-box;margin:0;padding:0;font-family:Arial}
body{background:#eef2f7;color:#1e293b}
header{background:#0f172a;color:white;padding:18px 30px;display:flex;justify-content:space-between}
nav a{color:#bfdbfe;text-decoration:none;margin-left:15px}

.hero{background:linear-gradient(120deg,#1d4ed8,#38bdf8);padding:35px;text-align:center;color:white}
.hero h2{margin-bottom:18px}

.search{background:white;padding:15px;border-radius:10px;display:flex;gap:10px;max-width:850px;margin:auto}
input,select{padding:10px;border:1px solid #ddd;border-radius:6px;flex:1;min-width:0}

button{background:#1d4ed8;color:white;border:0;padding:10px 16px;border-radius:6px;cursor:pointer}
.container{max-width:950px;margin:25px auto;padding:0 15px}
.card{background:white;padding:20px;border-radius:10px;margin-bottom:18px;box-shadow:0 3px 10px #ddd}
h3{color:#1d4ed8;margin-bottom:15px}

table{width:100%;border-collapse:collapse;font-size:14px}
th{background:#0f172a;color:white;padding:10px}
td{padding:10px;border-bottom:1px solid #ddd}
.status{color:green;font-weight:bold}

.details{display:flex;gap:10px}
footer{background:#0f172a;color:#bfdbfe;text-align:center;padding:15px}
</style>
</head>
<body>

<header><h2> 🚂 RailConnect</h2><nav><a href="#">Home</a><a href="#">Booking</a><a href="#">PNR
Status</a></nav></header>

<div class="hero"><h2>Plan Your Train Journey</h2>
```

```
<div class="search">
<input placeholder="From Chennai"><input placeholder="To Madurai">

<input type="date"><button onclick="alert('Searching trains...')">Search</button>
</div></div>

<div class="container">
<div class="card"><h3>Available Trains</h3>
<table>

<tr><th>Train</th><th>Departure</th><th>Fare</th><th>Status</th><th></th></tr>

<tr><td>Pandian Express</td><td>9:00 PM</td><td>₹420</td><td class="status">Available</td><td><button
onclick="alert('Train selected')">Book</button></td></tr>

<tr><td>Pearl City Express</td><td>8:30 PM</td><td>₹1120</td><td class="status">Available</td><td><button
onclick="alert('Train selected')">Book</button></td></tr>

<tr><td>Sethu Express</td><td>8:45 PM</td><td>₹650</td><td class="status">Available</td><td><button
onclick="alert('Train selected')">Book</button></td></tr>
</table></div>

<div class="card"><h3>Passenger Details</h3>
<div class="details"><input placeholder="Passenger Name"><input placeholder="Age"><input
placeholder="Mobile"></div><br>

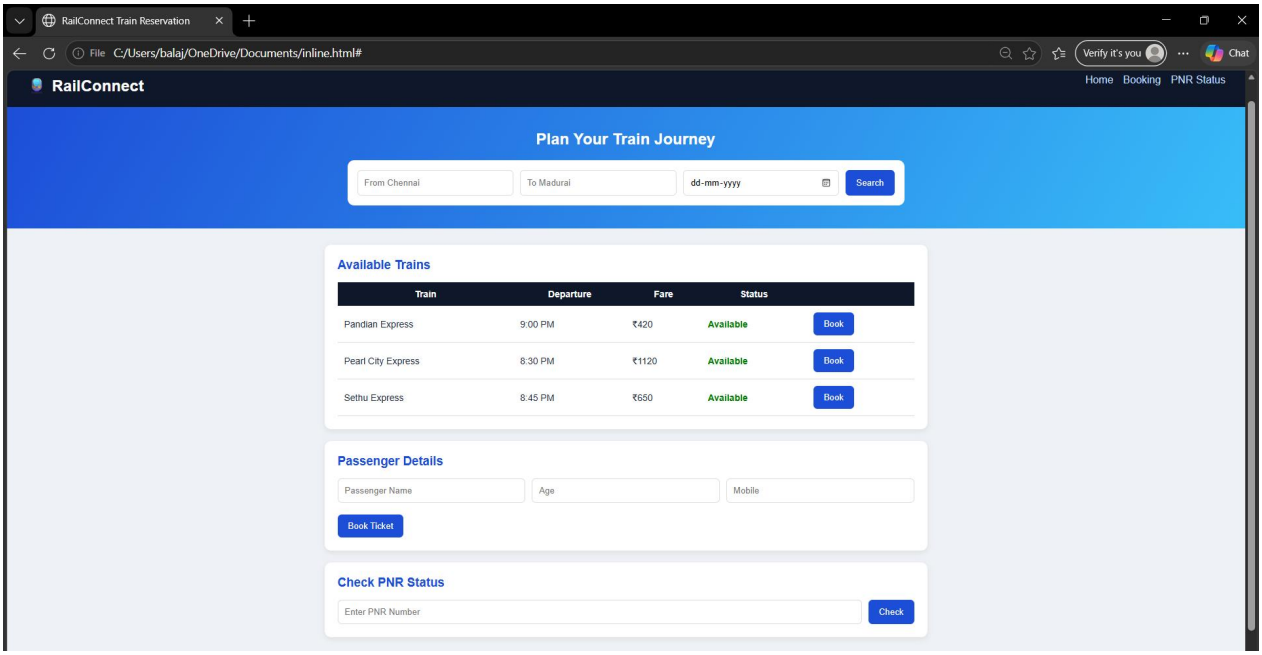
<button onclick="alert('Ticket booked successfully!')">Book Ticket</button></div>
<div class="card"><h3>Check PNR Status</h3>
<div class="details"><input id="pnr" placeholder="Enter PNR Number"><button
onclick="checkPNR()">Check</button></div>

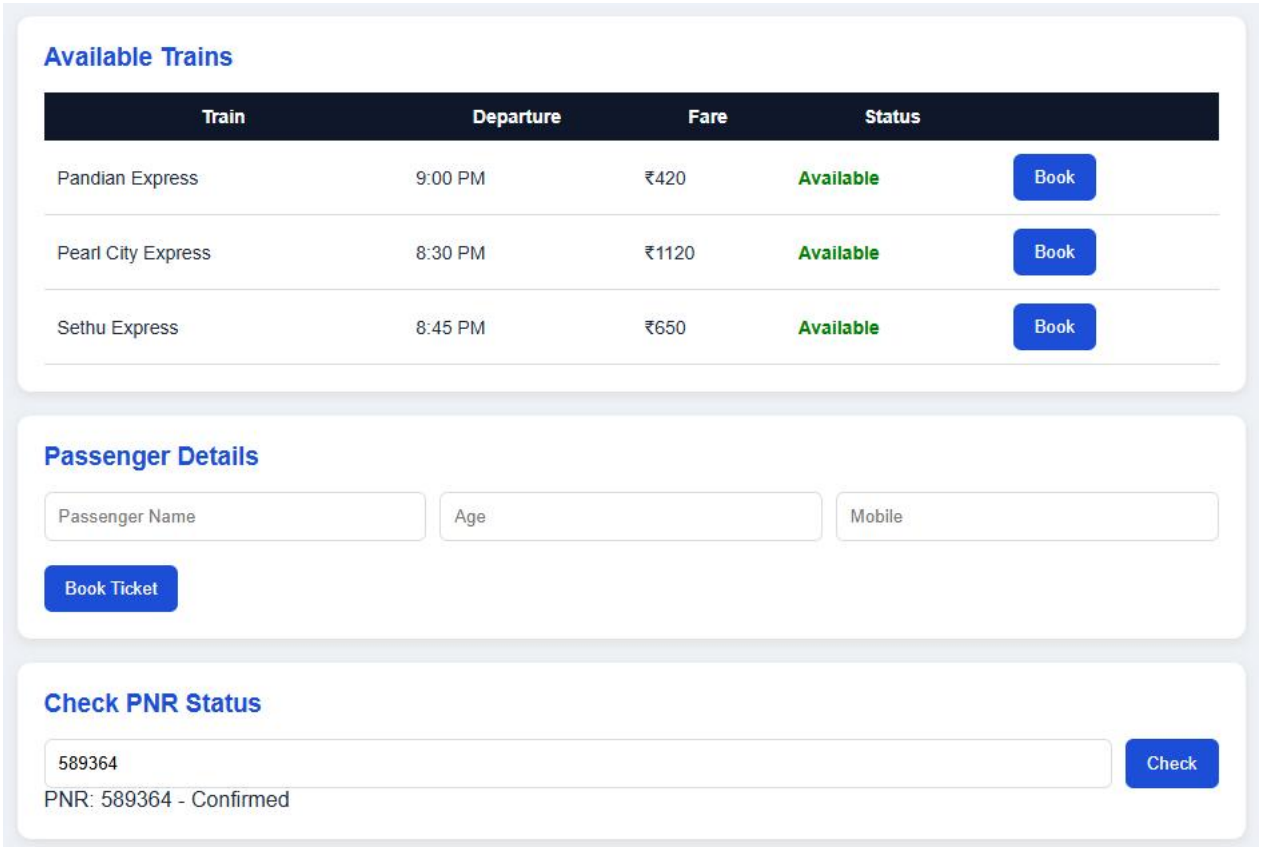
<p id="result"></p></div>
</div>
<footer>© 2026 RailConnect | Developed by Kirthanaa</footer>

<script>
function checkPNR(){
var p=document.getElementById("pnr").value;
document.getElementById("result").innerHTML=p?"PNR: "+p+" - Confirmed":"Please enter PNR";
}

</script>
</body>
</html>
```

Output:





Result:

Thus the Online Train Reservation System web application was designed, developed and executed successfully.

EXPERIMENT 5: ONLINE FLIGHT RESERVATION SYSTEM

Aim:

To build a responsive online flight reservation website using HTML, CSS and JavaScript with pages for admin login, flight search, ticket booking, payment, cancellation, feedback, booking status and ticket printing, with real-time seat availability and admin management of schedules and inventory.

Code (HTML, CSS and JavaScript):

```
<!DOCTYPE html>

<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">

<title>SkyWings Flight Reservation</title>

<style>
*{box-sizing:border-box;margin:0;padding:0;font-family:Arial}
body{background:#f0f9ff;color:#0f172a}

header{background:#0c4a6e;color:white;padding:18px 30px;display:flex;justify-content:space-between}
nav a{color:#bae6fd;margin-left:15px;text-decoration:none}
.hero{background:linear-gradient(120deg,#0369a1,#38bdf8);padding:35px;text-align:center;color:white}
.hero h2{margin-bottom:20px}

.search{background:white;padding:15px;border-radius:15px;display:flex;gap:10px;max-width:850px;margin:auto}
input,select{padding:10px;border:1px solid #ddd;border-radius:8px;flex:1;min-width:0}
button{background:#f97316;color:white;border:0;padding:10px 16px;border-radius:8px;cursor:pointer}
.container{max-width:950px;margin:25px auto;padding:0 15px}

.flight{background:white;padding:20px;border-radius:10px;margin-bottom:15px;display:flex;justify-content:space-
between;align-items:center;box-shadow:0 3px 10px #ddd}
.flight p{color:#64748b;margin-top:8px;font-size:14px}

.price{color:#f97316;font-size:22px;font-weight:bold;margin-bottom:10px}
.card{background:white;padding:20px;border-radius:10px;margin-top:20px;box-shadow:0 3px 10px #ddd}
h3{color:#0369a1;margin-bottom:15px}

footer{background:#0c4a6e;color:white;text-align:center;padding:15px;margin-top:30px}
</style>
</head>

<body>
<header><h2>✈ SkyWings Airlines</h2><nav><a href="#">Home</a><a href="#">Flights</a><a href="#">My
Ticket</a></nav></header>
```

```
<div class="hero"><h2>Fly Anywhere, Anytime</h2>
<div class="search"><input placeholder="From (MAA)"><input placeholder="To (DEL)"><input type="date"><button
onclick="alert('Searching flights...')">Search</button></div></div>
<div class="container">
<div class="flight"><div><h3>IndiGo Airlines</h3><p>06:10 AM - 08:55 AM | Non-stop</p></div><div><div
class="price">₹5,480</div><button onclick="alert('Flight selected!')">Book</button></div></div>

<div class="flight"><div><h3>Air India</h3><p>11:30 AM - 02:20 PM | Non-stop</p></div><div><div
class="price">₹6,120</div><button onclick="alert('Flight selected!')">Book</button></div></div>
<div class="flight"><div><h3>Vistara Airlines</h3><p>06:45 PM - 09:30 PM | Non-stop</p></div><div><div
class="price">₹7,350</div><button onclick="alert('Flight selected!')">Book</button></div></div>

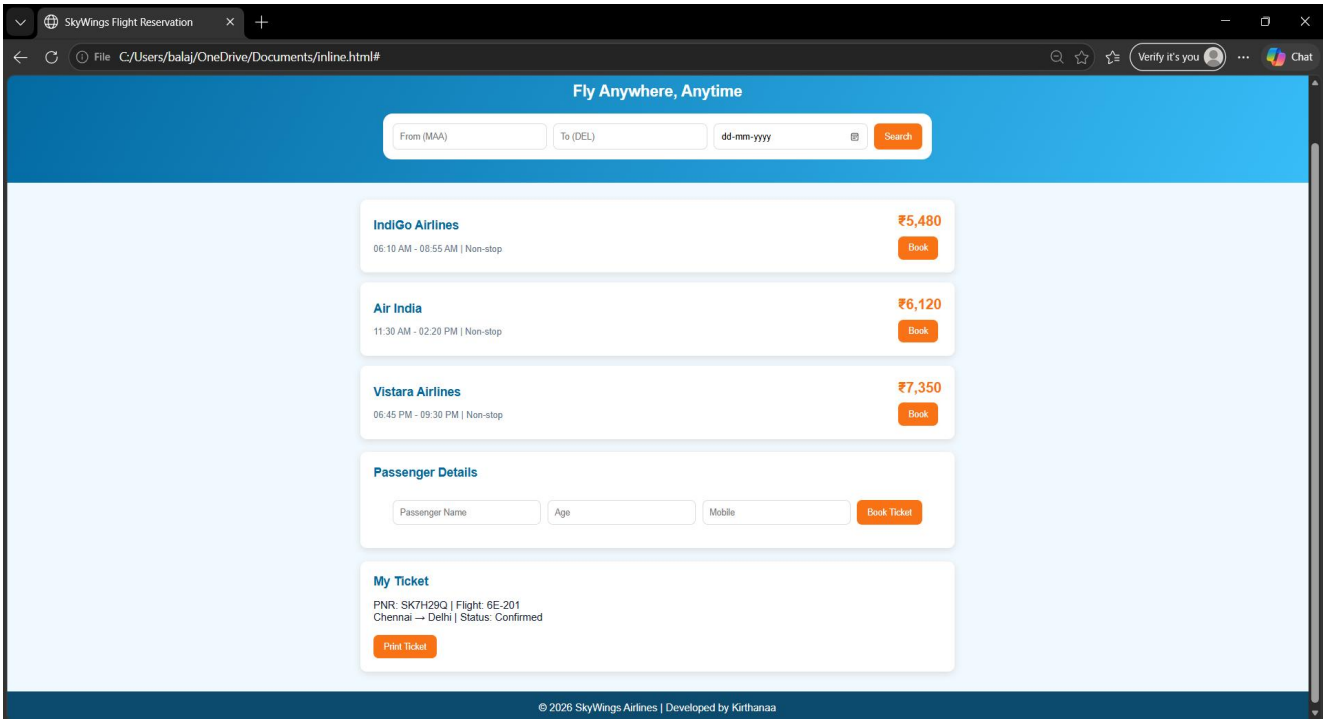
<div class="card"><h3>Passenger Details</h3>
<div class="search"><input placeholder="Passenger Name"><input placeholder="Age"><input
placeholder="Mobile"><button onclick="alert('Ticket booked successfully!')">Book Ticket</button></div></div>

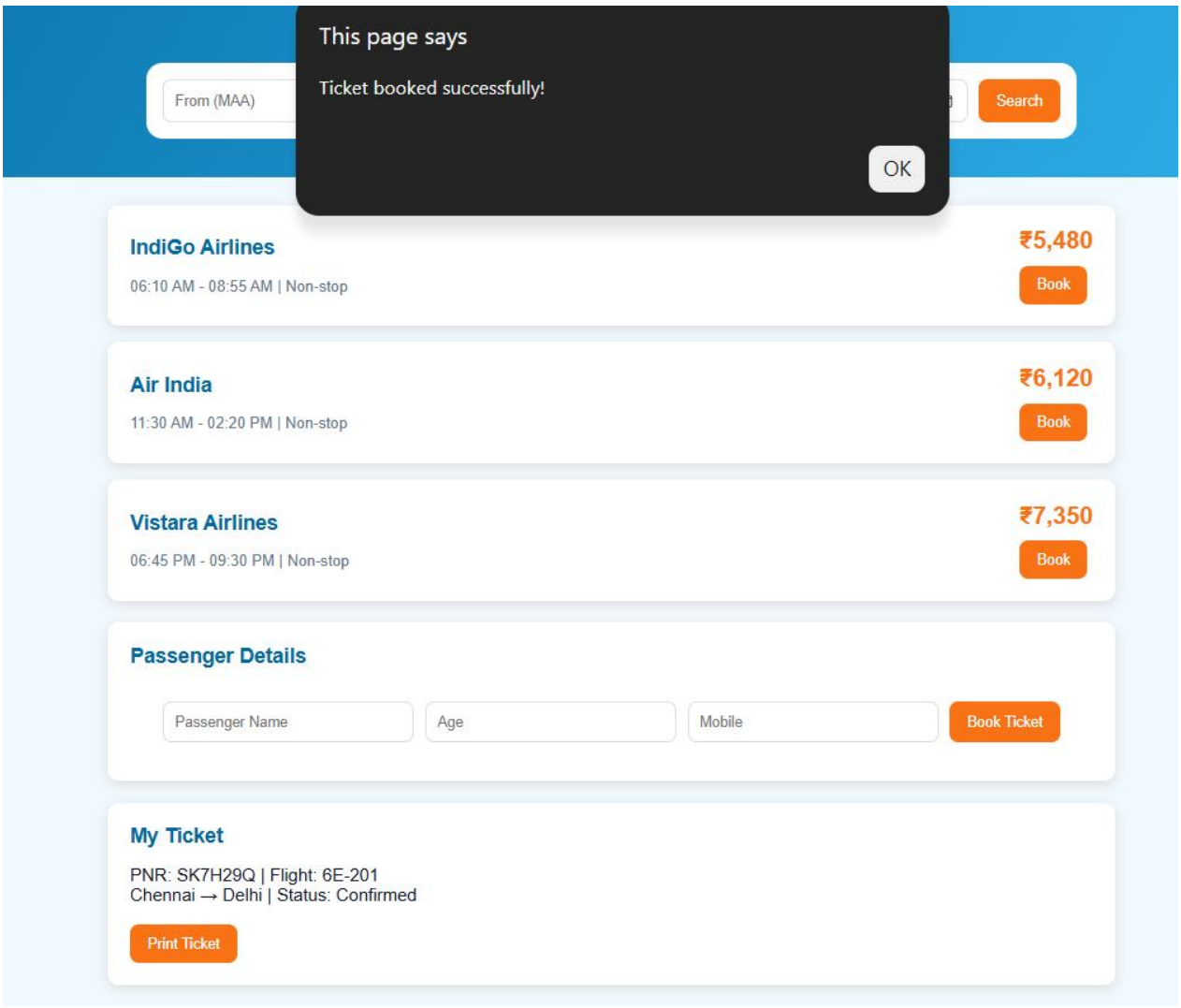
<div class="card"><h3>My Ticket</h3><p>PNR: SK7H29Q | Flight: 6E-201</p><p>Chennai → Delhi | Status:
Confirmed</p><br><button onclick="window.print()">Print Ticket</button></div>
</div>

<footer>© 2026 SkyWings Airlines | Developed by Kirthanaa</footer>

</body>
</html>
```

Output:





Result:

Thus the Online Flight Reservation System web application was designed, developed and executed successfully.